

# EE 122/ME 141 Week 10, Lecture 2: Examples of state-feedback control design

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**Example 1.** Consider the first-order system

$$\dot{x} = -0.5x + u, \quad y = 2x.$$

**Task 1.** Write  $A$ ,  $B$ ,  $C$ , and  $D$  matrices for the system.

**Task 2.** Note the open-loop pole(s) of the system.

**Task 3.** Physically, we want the closed-loop output  $y(t)$  to reach 63% of its final value in 1 second when a step input is applied. What is your desired pole?

**Task 4.** Find the state-feedback controller  $K$  and the feedforward gain  $k_f$

$$u = -Kx + k_fr$$

that achieves your desired step response.

**Example 2.** Consider the DC motor transfer function

$$G(s) = \frac{2}{2s + 1}.$$

**Task 1.** Write a state-space representation in reachable canonical form.

**Task 2.** Note down  $A$ ,  $B$ ,  $C$ , and  $D$ .

**Task 3.** Note the open-loop pole(s) of the system.

**Task 4.** Physically, we want the closed-loop output  $y(t)$  to reach 63% of its final value in 1 second when a step input is applied. What should the desired pole be?

*Hint:* For a first-order closed-loop system

$$G_{cl}(s) = \frac{b}{s + a},$$

a unit step input gives

$$Y(s) = \frac{b}{s(s + a)},$$

so

$$y(t) = y(\infty) (1 - e^{-at}).$$

**Task 5.** Find the state-feedback controller

$$u = -Kx$$

that achieves your desired pole.

**Example 3.** Consider the second-order system in reachable canonical form

$$\dot{x} = Ax + Bu, \quad y = Cx + Du$$

with

$$A = \begin{bmatrix} -2 & -13 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad D = 0.$$

**Task 1.** Run the open-loop step response in MATLAB:

$$\text{step}(\text{ss}(A,B,C,D))$$

What do you observe about the oscillation frequency of the system?

**Task 2.** Note the open-loop poles of the system.

*MATLAB hint:*

$$\text{eig}(A)$$

**Task 3.** We want the closed-loop system to have a lower oscillation frequency than the open-loop system. Propose a pair of desired poles.

**Task 4.** Use MATLAB to compute the state-feedback gain:

$$K = \text{place}(A,B,DP)$$

where DP is your vector of desired poles.

**Task 5.** With state feedback  $u = -Kx$ , write the closed-loop matrix

$$A_{cl} = A - BK$$

in terms of

$$K = \begin{bmatrix} k_1 & k_2 \end{bmatrix}.$$

**Task 6.** Compare the entries of  $A$  and  $A_{cl}$ . Which coefficients change under state feedback? How does this connect to what we saw in controller canonical form in class?